

A small module that can:

1. Take **current job role** (e.g., “Data Analyst”)
2. Recommend **next possible roles** (e.g., “Senior Data Analyst”, “BI Analyst”, “Data Scientist”)
3. For each next role, recommend **skills to learn** + priority
4. Explain “why” (skill gaps)

Module design

1) Inputs

- `current_role` (required)
- `current_skills` (optional list)
- `experience_level` (optional: junior/mid/senior)
- `domain` (optional: finance, healthcare, etc.)

2) Outputs

For each recommended next role:

- `next_role`
- `match_score` (0–1 or 0–100)
- `missing_skills` (top 5–15)
- `recommended_learning_order`
- `reason` (short explanation)

Data you NEED

Create these 3 tables (CSV/JSON is fine):

A) Role → Skills matrix (most important)

Each role has skills with importance.

- `role_name, skill_name, importance(1-5)`

Example:

- Data Analyst → SQL (5), Excel (4), PowerBI (4), Statistics (3)

B) Skill taxonomy (optional but good)

- `skill_name, category, synonyms`

Example: "Power BI" = "pbi", "powerbi"

C) Career transitions (optional but powerful)

- `from_role, to_role, likelihood_weight`

Example: Data Analyst → Senior Data Analyst (0.9), Data Analyst → Data Scientist (0.4)

Recommendation Logic (simple but correct)

You have 2 solid options. Pick one based on time.

Option 1: Similarity-based next roles (fastest)

1. Convert each role into a "skill vector"
 - Each skill is a feature, value = importance
2. Compute similarity between current role and all roles

- cosine similarity works

3. Recommend top N roles with similarity

Then skill gap:

- Missing skills = skills required by target role but not in current role / user skills
- Rank missing skills by importance

✓ This is easy to justify and implement.

Option 2: Graph-based career path (better “career path” feel)

Use transitions table:

- From current role → suggest top next roles by likelihood
- If you want 2-step path: also show next-next roles (like a roadmap)

Architecture (clean “module” separation)

Do this structure (even if it's one repo):

- `data/roles_skills.csv`
- `src/recommender.py` ← core logic
- `src/api.py` (optional FastAPI endpoint)
- `demo.ipynb` (show results)

```
{
  "current_role": "Audit Associate",
  "recommended_roles": [
    {
      "role": "Senior Audit Associate",
      "match_score": 0.82,
      "missing_skills": ["Advanced Auditing", "Risk Analysis"]
    },
    {
      "role": "Audit Manager",
      "match_score": 0.64,
      "missing_skills": ["Leadership", "Regulatory Strategy"]
    }
  ]
}
```

